H

millennia institute

Class: Adm No

Candidate Name: _____

2026 Preliminary Examinations
Pre-University 3

GEOGRAPHY (HIGHER 2) **9173/02**
Paper 2 **14 September 2026**

3 hours

Additional Materials: Insert, Answer Booklet(s)

READ THESE INSTRUCTIONS FIRST

An answer booklet will be provided with this question paper. You should follow the instructions on the front of the answer booklet. If you need additional writing paper, ask the invigilator for a continuation booklet.

Section A
Answer **both** questions.

Section B
Answer **one** question.

The number of marks is given in brackets [ ] at the end of each question or part question.

Section A

Answer **both** questions.

Cluster 4 Fieldwork

1 A group of 15 students decided to investigate the local community's response to climate change in Rotterdam, the Netherlands. Rotterdam is a low-lying city that is increasingly vulnerable to intense rainfall and pluvial flooding. Various measures, including water squares, green spaces, permeable surfaces and stormwater-storage facilities, have been introduced to improve the city's resilience.

The students selected five locations in Rotterdam: Water Square Benthemplein, Rotterdam Centraal and Kruisplein, Noordplein, Dakpark Rotterdam and Museumpark.

The students agreed to investigate the following hypothesis:

The extent of public climate-adaptation measures implemented by the Municipality of Rotterdam is greater than the extent of household climate-adaptation measures adopted by individuals in Rotterdam.

The students conducted secondary research online for basic information about the five locations and the climate-adaptation measures found at each location.

They designed a questionnaire to investigate people's awareness of, support for and adoption of climate-adaptation measures. They also produced an environmental data-collection sheet to record the climate-adaptation features observed at each location.

The students used convenience sampling to survey 25 people at each location. They conducted their fieldwork from 11 am to 1 pm on two weekdays in October. It was cool and rainy on both days. In total, the students surveyed 125 people.

At each location, the students carried out the following steps:

- They approached people who appeared to be at least 18 years old.
- They verified whether each person lived, worked or studied in Rotterdam.
- They asked willing participants to complete the questionnaire.
- They identified climate-adaptation features within 50 metres of the survey point.
- They used the environmental data-collection sheet to give each feature a score based on its perceived effectiveness in managing intense rainfall.

Resource 1 shows the locations of the five fieldwork sites in Rotterdam. Resource 2 shows basic information about the five locations and their climate-adaptation measures. Resource 3 is a photograph of Benthemplein Water Square during dry and wet weather. Resource 4A shows the survey questionnaire used by the students. Resource 4B shows the environmental data-collection sheet used by the students. Resource 5 shows the results collected for environmental perception survey during the investigation.

(a) Explain the suitability of the students' hypothesis for investigating the Rotterdam community's response to climate change. [4]

(b) Explain the limitations of the sampling method used by the students. [4]

(c) With reference to Resources 1 and 3, explain how the students could minimise potential risks when undertaking the fieldwork. [6]

(d) Using Resources 4A and 4B, explain the strengths and limitations of the survey questionnaire and environmental perception survey. [6]

(e) Explain how students could improve the accuracy and reliability of their data collection procedures during the geographical investigation. [5]

(f) Suggest how the 'mean environmental perception scores for Museumpark' shown in Resource 5 may be represented. [5]

(g) Using all the Resources, evaluate the validity of the data collected in determining whether the students' hypothesis about public implementation and household adoption is supported. [10]

Cluster 3 Sustainable Future and Climate Change

2 Resource 6 shows the average carbon dioxide emissions per capita across the world in 2019. Resource 7 shows the share of plastic waste management methods of Europe and Asia in 2019. Resource 8 shows a graph of atmospheric CO_2 (parts per million, NOAA) and Global Temperature Anomaly (°C, GISS) from 1964 to 2008. Resource 9 is a photograph of an aerial view of the Bengal Delta in Bangladesh in April 1991.

(a) Using Resource 6, suggest reasons for the variations in the methods of plastic waste management between Europe and Asia in 2019. [6]

(b) Using Resource 7, explain the impact of multiple deprivations experienced by slum dwellers on sustainable urban development. [6]

(c) Citing data from Resource 8, describe the relationship between changes in atmospheric CO_2 concentration and global temperature anomaly over time. [4]

(d) Explain reasons for uncertainty over the present and future impacts of contemporary climate change. [4]

(e) Using Resource 9, explain the impacts of coastal flooding on different groups of people and the environment in Bangladesh. [4]

(f) With the use of examples, explain the impacts of changes in aquatic ecosystems brought about by climate change on humans. [6]

Section B

Answer **either** Question 3 **or** Question 4.

Cluster 3 Sustainable Future and Climate Change

3 'Liveability is more complementary than it is in tension with sustainable urban development.'

Evaluate this statement. [20]

4 "Human activities, principally through emissions of greenhouse gases, have unequivocally caused global warming."

Evaluate this statement. [20]

End of Paper

Copyright Acknowledgements

Resource 1	Google maps (author's own)
Resource 2	https://landezine.com/water-square-benthemplein-by-de-urbanisten/ (Last accessed 26 July 2026)
Resource 3	https://www.urbanisten.nl/work/benthemplein https://www.iurc.eu/wp-content/uploads/2022/05/Rotterdam-Presentation_compressed.pdf https://www.rotterdam.info/en/visit/finder-locations/rotterdamse-oogstmarkt https://www.rotterdam.info/en/visit/finder-locations/dakpark https://paulderuiter.nl/en/projects/parkeergarage-museumpark (Last accessed 1 August 2026)
Resource 4A and 4B	Author's own
Resource 5	Author's own
Resource 6	https://ourworldindata.org/grapher/share-plastic-fate?time=2019 (Last accessed 19 August 2026)
Resource 7	https://www.justonewayticket.com/smokey-mountain-a-walk-through-the-slums-of-manila-philippines/ (Last accessed 19 August 2026)
Resource 8	https://skepticalscience.com/the-correlation-between-co2-and-temperature.html (Last accessed 19 August 2026)
Resource 9	https://www.lse.ac.uk/granthaminstitute/publication/tackling-flooding-in-bangladesh-in-a-changing-climate/ (Last accessed 19 August 2026)